

Lecture 4: Paging

Fundamentals of Operating System Design (CEUC301)

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Learning Outcomes

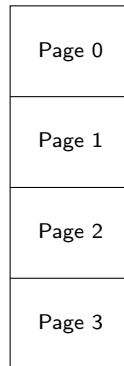
- Explain paging.
- Define pages, frames and page tables.
- Perform logical to physical address translation.
- Explain page table structure.

Why Paging?

- Eliminates external fragmentation.
- Enables non-contiguous allocation.
- Simplifies memory management.
- Forms the basis of virtual memory.

Basic Concept

- Logical memory is divided into pages.
- Physical memory is divided into frames.
- Page size = Frame size.



Address Translation

Logical Address = Page Number + Offset

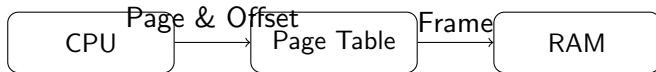
Physical Address = Frame Number + Offset

Page Table

- Maps each logical page to a physical frame.
- Each process has its own page table.
- Maintained by the operating system.

Page No.	Frame No.
0	5
1	2
2	8
3	1

Paging Address Translation



Offset remains unchanged during translation.

Address Translation Example

Given:

- Page size = 1 KB
- Logical address = 2500
- Page table: Page 2 $\rightarrow$ Frame 8

$$2500 = (2 \times 1024) + 452$$

Therefore,

- Page Number = 2
- Offset = 452
- Physical Address = $(8 \times 1024) + 452 = 8644$

Practice Problem

Page Size = 512 Bytes

Logical Address = 1450

Page Table

Page	Frame
0	4
1	7
2	3
3	9

Find:

- 1 Page Number
- 2 Offset
- 3 Physical Address

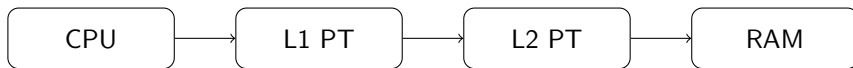
Page Table Entry (PTE)

Typical fields in a Page Table Entry:

- Frame Number
- Valid / Invalid Bit
- Protection Bits (R/W/X)
- Reference Bit
- Dirty (Modified) Bit

Multi-level Paging

- Single page tables become very large.
- Divide the page table into multiple levels.
- Allocate second-level tables only when required.
- Reduces memory overhead.



Advantages of Paging

- Eliminates external fragmentation.
- Supports virtual memory.
- Efficient memory allocation.
- Enables memory protection.
- Allows sharing of pages.

Disadvantages of Paging

- Internal fragmentation may occur.
- Additional memory for page tables.
- Address translation overhead.
- TLB required for high performance.

Comparison: Contiguous Allocation vs Paging

Feature	Contiguous Allocation	Paging
Allocation	Continuous block	Non-contiguous frames
External Fragmentation	Yes	No
Internal Fragmentation	Possible (fixed partitions)	Possible (last page)
Address Translation	Simple	Uses Page Table
Memory Utilization	Lower	Better
Virtual Memory Support	Limited	Excellent

Numerical Example

Given:

- Logical Address Space = 64 KB
- Page Size = 4 KB

Find:

- ① Number of pages
- ② Offset bits
- ③ Page number bits

Answer:

$$\text{Pages} = \frac{64}{4} = 16$$

Offset bits = 12

Page bits = 4

Summary

- Paging divides logical memory into pages.
- Physical memory is divided into frames.
- Page tables perform page-to-frame mapping.
- Multi-level paging reduces page table size.
- Paging eliminates external fragmentation.

Questions

- 1 Explain paging with a neat diagram.
- 2 Explain logical to physical address translation.
- 3 Explain the structure of a page table.
- 4 What is multi-level paging?
- 5 Solve a paging numerical problem.